that their routers offer. In fact, ASUS's firmware in their mid to high-end routers is based on open-source firmware called ASUS-WRT.

Coming back to the matter at hand, the routers that we received were quite feature-rich. However, we're pretty sure that most people will not even look beyond compatibility with their Internet connection and the speed on offer. Especially for such users, Belkin has one of the most beginner friendly UIs we've seen so far. It's like looking at one of those textbooks that you'd have in kindergarten. You simply cannot make a mistake.

D-Link, ASUS, Netgear and Trendnet have interface that are for those of you who can configure their own routers and don't need the extra help. TP-LINK has maintained the age-old familiar interface and we still appreciate the simplicity and comfort of having all configurable options on the left column. Navigating through TP-LINK's interface is the fastest since all changes made are saved temporarily and do not come into effect until you reboot the router. Most other manufacturers have a mini-reboot each time you modify any parameter and it can be quite painful when you have to set up over 20 routers in a day.

Digisol has some good products that work wonderfully and provide a stable bandwidth under full load conditions, however, their interface on entry-level routers needs a lot of work. Since it's an Indian brand very few open-source volunteers can get their hands on these devices and hence the prospect of seeing an open-source firmware on these devices is far-fetched.

HOW WE TESTED

1. Preparation

Creating an ideal test environment in our office was near impossible, but we did our best to ensure a minimum of interference from outside sources. We disabled as many unnecessary routers as we could and even the essential ones were switched to other channels. As a real world environment at least, we were all set.

Each router was assembled and connected to the Internet for firmware upgrades if required. Certain routers wouldn't get past the QIS screen unless connected to an RJ11 jack and those we promptly dropped as we didn't have a DSL line to spare. After upgrades, the routers were hooked up to a test rig for initial setup and individually tested as required.

To maintain a fair playing field, all routers were switched to mixed mode a/b/g/n/ac mode and open authentication was used. These routers were not connected to the internet during the testing phase so as to avoid unnecessary loss in performance. Also the orientation of the router was adjusted to ensure maximum signal strength at both client locations.

Two machines were used – One was connected using a CAT6 ethernet cable directly to the router's LAN port while the other was connected wirelessly at a fixed range. Each machine connected was pretty much top of the line with more than enough read and write speeds owing to 2x SSDs in a RAID 0 configuration. Needless to say there wouldn't be any bottlenecks at these locations.

Machine 1 (Client) Configuration:
Processor: Intel Core i7 3770K
Motherboard: ASRock Z77 Extreme 4
Primary drive: OCZ Vector 256 GB
RAM: Kingston HyperX 2x4GB @ 2133 MHz
Endpoint: ASUS PCE-AC68
Endpoint: Trendnet TEW-805UB
SMPS: Corsair RM650
Cabinet: Cooler Master HAF-XM

Machine 2 (Server) Configuration:
Processor: Intel Core i7 4770K
Motherboard: ASRock Z87 Extreme 6
Primary drive: Kingston HyperX 2x120GB SSD in RAID 0
RAM: ADATA 4x4GB @ 2333 MHz
SMPS: Cooler Master 850W

2. Endpoints:

The endpoints used on the client side was primarily the ASUS PCE-A68 owing to the ridiculous bandwidth it provided on all three channels. TurboQAM modulation was enabled and the antennae were adjusted according to the recommended specifications. In case the ASUS PCIE WLAN card didn't connect to a particular router then the Trendnet TEW-805UB was used. The Trendnet endpoint was used at multiple locations to simultaneously fill up the router bandwidth. We made use of two TEW-805UB for this purpose.

3. Tools used

We wished to ensure that the software that we used could be used by anyone reading the article

The Tamosoft Throughput tool was used to flood the router with both TCP and UDP traffic to figure out how the default QoS settings worked. LST Speed Test and LST Server were used to replicate the same results. In Zone2 we faced a little problem regarding UDP traffic, almost all packets were getting dropped in Tamosoft and so we used iperf to test TCP and UDP traffic in Zone2. There was a little hassle involved in the same given that Linux drivers for the ASUS PCE-A68 were yet to be released. Hence, we used the windows drivers encapsulated with the ndiswrapper to test throughput. Needless to say, TCP bandwidth was given greater weightage than UDP bandwidth.

WirelessMon was used to monitor the Wi-Fi signal strength. Pinging the machines was done via the command prompt.

For file transfer tests were performed on both the 2.4GHz and 5GHz bands. A single 100MB file and an assorted collection of files totalling 100MB were used at 2.4GHz; a single 1GB file and an assorted collection of files totalling 1GB were used at 5GHz.

For HD media streaming tests, we used a true 1080p movie compressed to mkv and streamed via VLC using TCP. All playback above 12500000bps on the client was considered a stutter free experience.

Performance testing was purely objective in nature while feature scoring was subjective as always. The routers were, obviously, classified into single-band and dual-band categories for this test.

Build quality

They being routers, we didn't really expect to see metallic bodies, especially on those with internal antennae, but we did expect some to see some robust plastic shells at the very least. In the latter department we were a bit disappointed, with only the Edimax BR-6478AC, Digisol DG-BR4000N\E and the Netgear JNR3210 having a tough exterior and of heft to them, (sigh).

Almost all high-end routers had good build quality though and seemed capable of taking a few falls. The one issue here